

Is VPN Really Dead and Replaced by Zero Trust Network Access (ZTNA)?

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Cisco Webex App

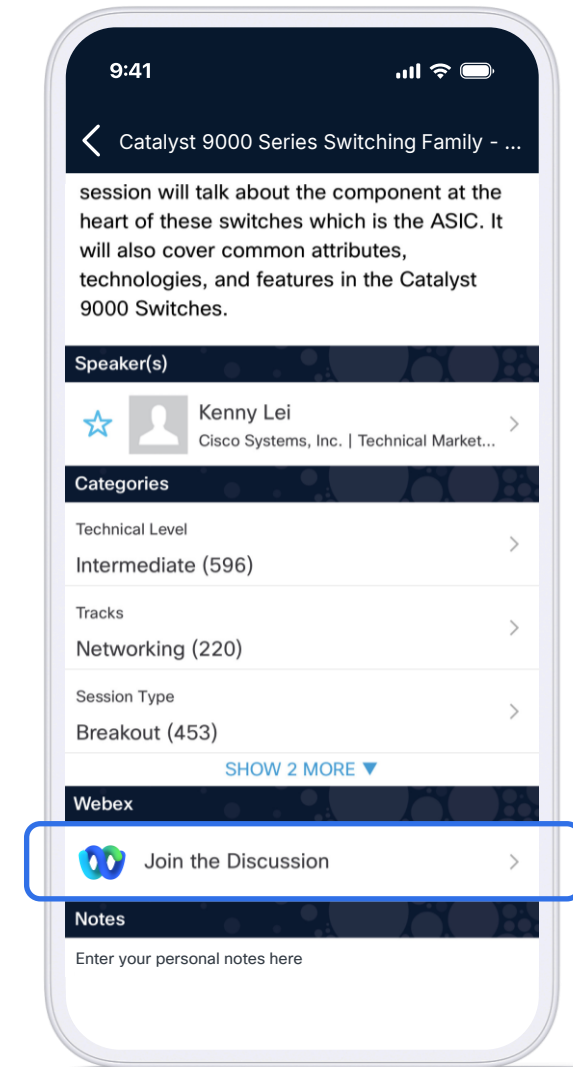
Questions?

Use Cisco Webex App to chat with the speaker after the session

How

- 1 Find this session in the Cisco Live Mobile App
- 2 Click "Join the Discussion"
- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until June 13, 2026.



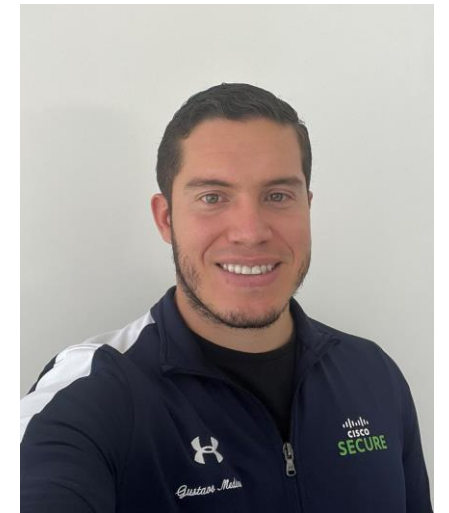
<https://ciscolive.ciscoevents.com/clamer26/BRKSEC-1015>

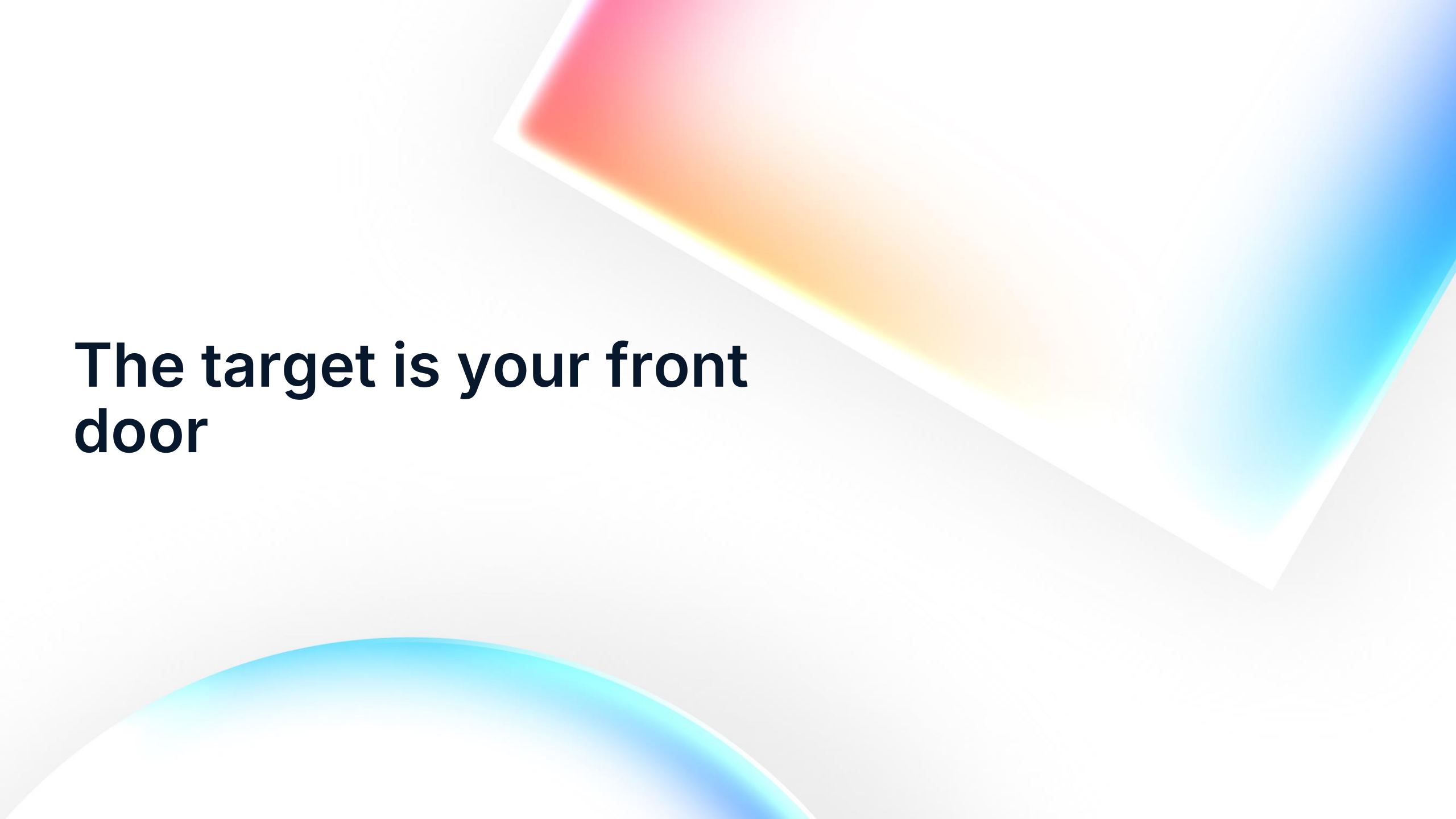
Agenda

- 01 Introduction**
- 02 What is ZTNA?**
- 03 VPN vs ZTNA: Key Differences**
- 04 VPN Common Objections**
- 05 Why ZTNA?**
- 06 Universal ZTNA (UZTNA)**
- 07 Cisco Secure Access**
- 08 Flexible Journey to ZTNA**

\$ whoami

- Gustavo Medina
- Technical Solutions Architect
- Costa Rican cr
 - Currently living in Mexico mx
- Joined Cisco (TAC) in 2009
- CCIE Security #51487
- Football Fan

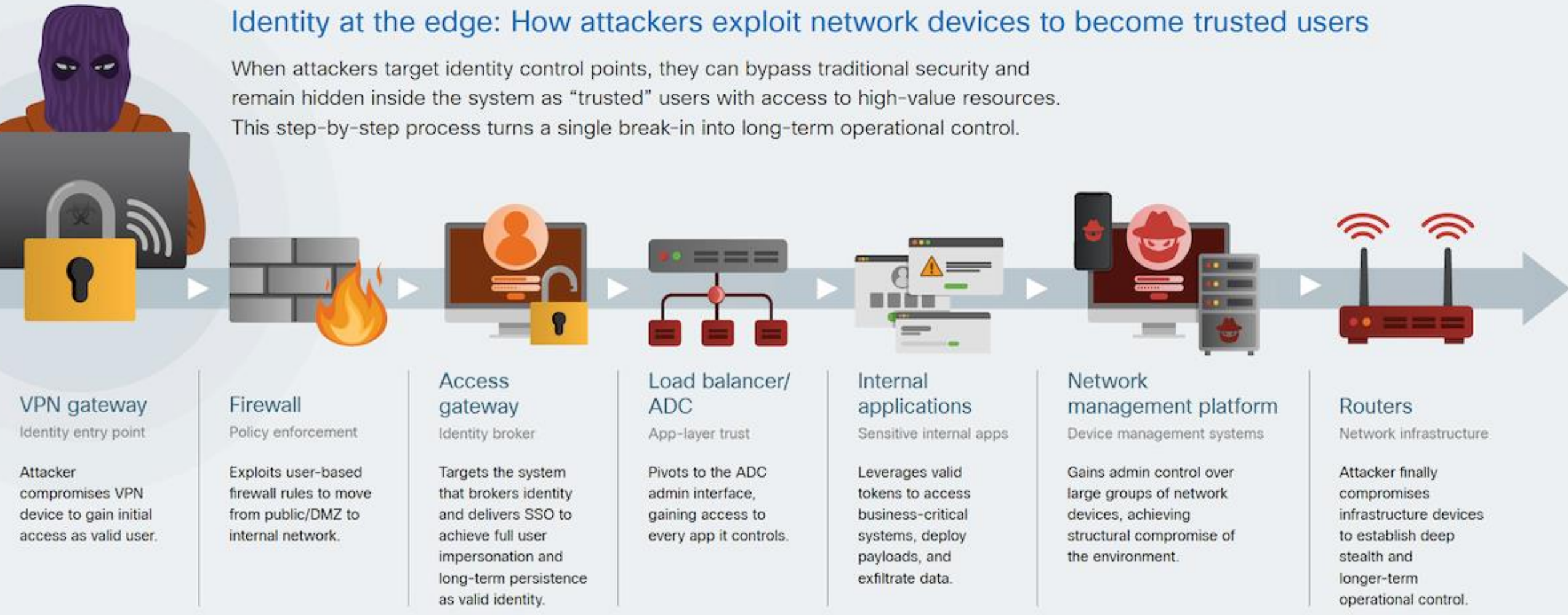




**The target is your front
door**

Identity at the edge: How attackers exploit network devices to become trusted users

When attackers target identity control points, they can bypass traditional security and remain hidden inside the system as “trusted” users with access to high-value resources. This step-by-step process turns a single break-in into long-term operational control.



Talos 2025 Year in Review:

most-targeted vulnerabilities shows nearly a quarter landing on network infrastructure

The target is your front door

- Ivanti Connect Secure: [CVE-2024-21887](#) was the headline, part of an in-the-wild zero-day pattern running 2023 through 2025
- Fortinet: [CVE-2024-55591](#) FortiOS authentication bypass, exploited before public disclosure.
- Half a dozen more across the same window. The pattern is **not vendor specific**.

Yes, Cisco is on that list



April 2024: State-sponsored actor UAT4356 (Storm-1849) targeted Cisco ASAs and FTDs.

September 2025: Fresh wave assessed with high confidence as the same actor.

April 2026: Threat actor developed a previously unknown persistence mechanism that was preserved across upgrading to the fixed releases that were published in September 2025

Exploitation of vulnerabilities is now the most common initial access vector for breaches

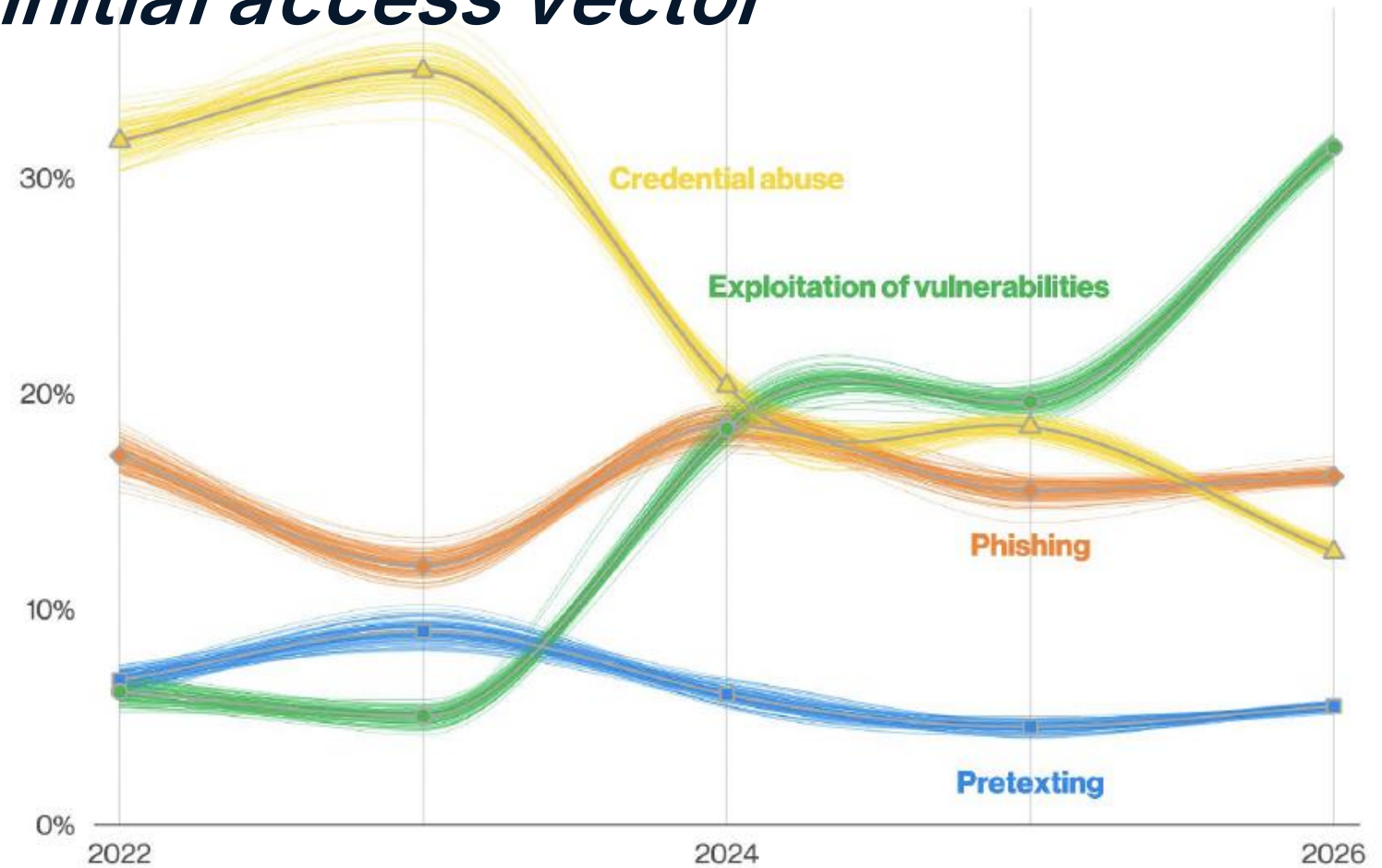


Figure 5. Known initial access vectors in non-Error, non-Misuse breaches over time (n for 2026 dataset=19,905)

<https://www.verizon.com/business/resources/reports/2026-dbir-data-breach-investigations-report.pdf>

The slide where ZTNA magically replaces everything is the part where the architecture starts lying.

When a vendor walks into your office and tells you “VPN is dead”.
Your gut says they are right. The slide they hand you next, the
one where ZTNA magically replaces everything, is the part where
the architecture starts lying to you.

Today, you walk out knowing the difference.

"VPN is dead" is a sales pitch, not an architecture

VPN is a connectivity protocol stack.

Zero Trust is a security architecture and operating model.

A protocol does not "die" when a better model emerges.

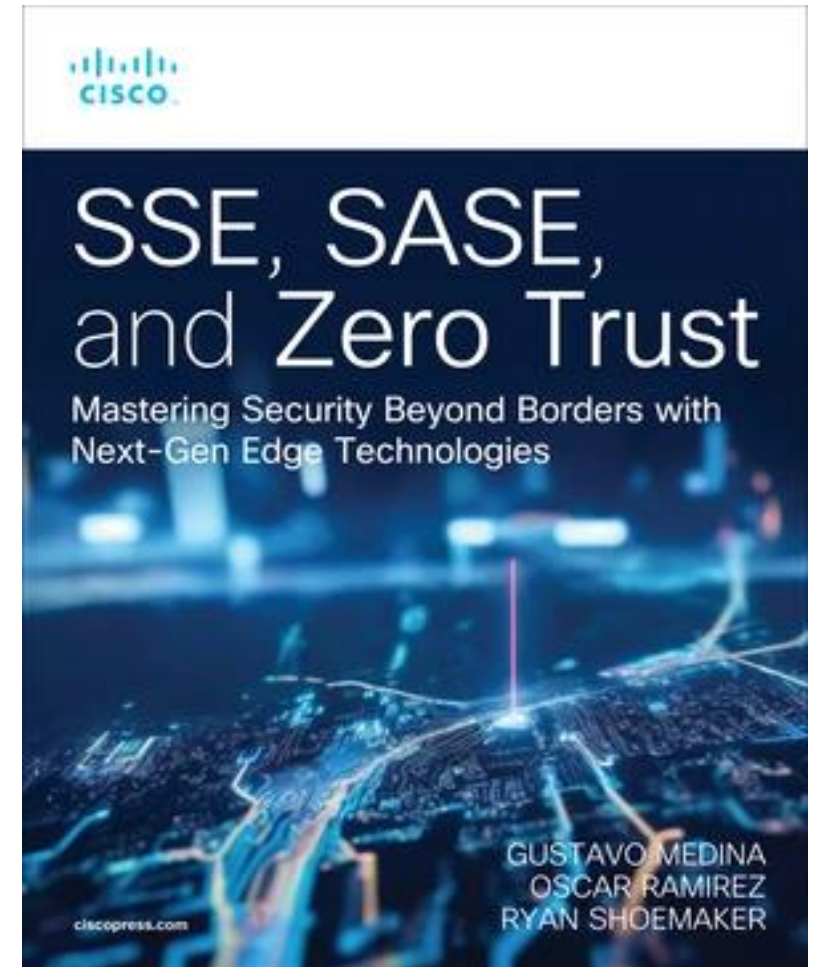
It gets complemented, constrained, or replaced depending on the use case.

The architect's question is not: **"Is VPN dead?"**

It is:

"Which access pattern belongs where, under which trust assumptions, with which enforcement model, and why?"

VPN may still fit certain network-level access needs.



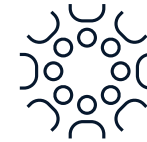
The background features a light gray gradient. A white rectangle is tilted diagonally from the top right towards the bottom left. A vibrant rainbow gradient, transitioning from red to orange, yellow, and blue, is applied to the rectangle. In the bottom left corner, there is a large, semi-circular shape with a blue-to-white gradient.

What is ZTNA?

Gartner



**Internal Workforce
Remote Access**



**Privileged
Remote Access**



**Extended Workforce &
BYOD**



On-prem Access

VPN vs ZTNA

Key Differences

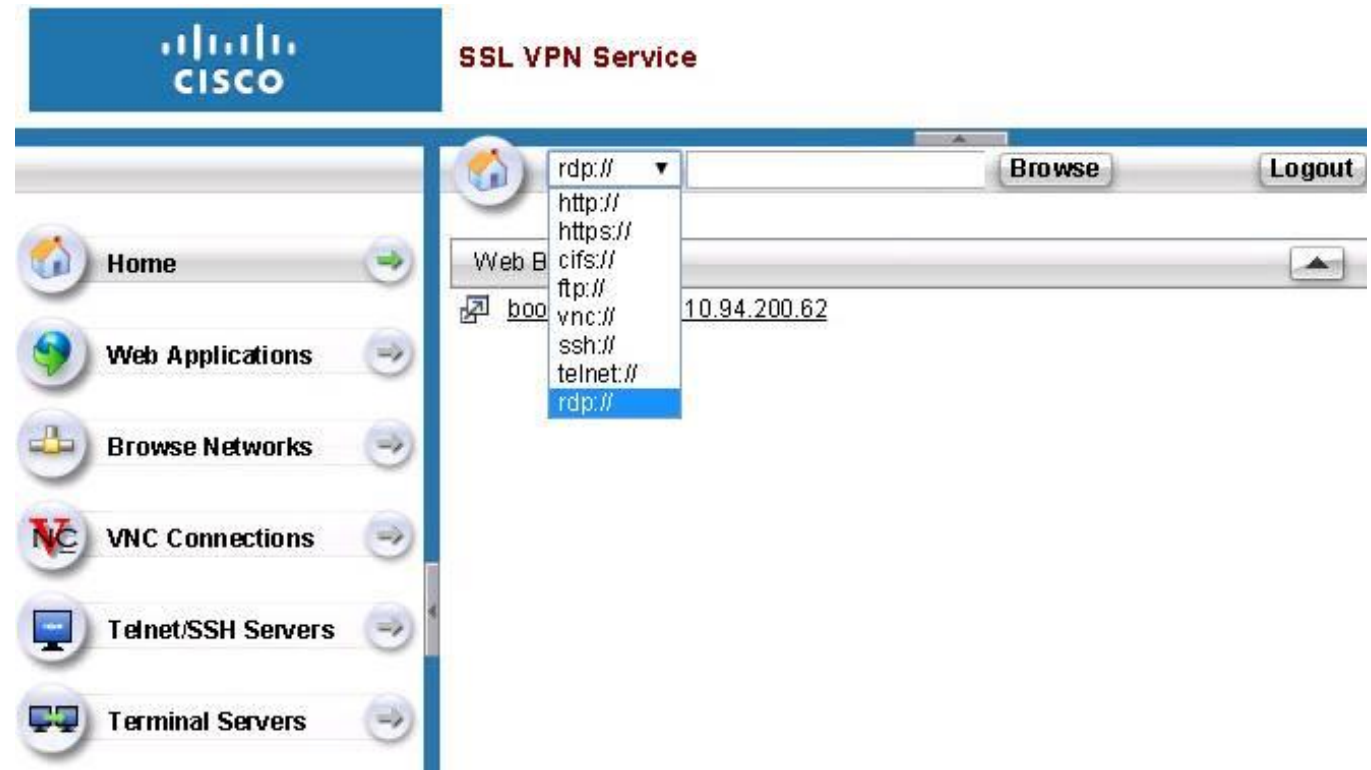
	VPN	ZTNA
User Experience	Requires VPN client software	No client software required *
Access Scope	Access to full network or network segment	Access to specific applications
Posture Checks	Posture assessed once at VPN authentication	Posture assessed at each application access
Connection Architecture	1:1. Client-to-Headend relationship	Client can connect to different headends per application
Trust Model	Implicit	Explicit

We had WebVPN Clientless before ZTNA was even a concept



Supported since ASA 7.1
*Deprecated on 9.17

VPN 3000
Series Concentrator
supported Clientless



VPN Common Objections

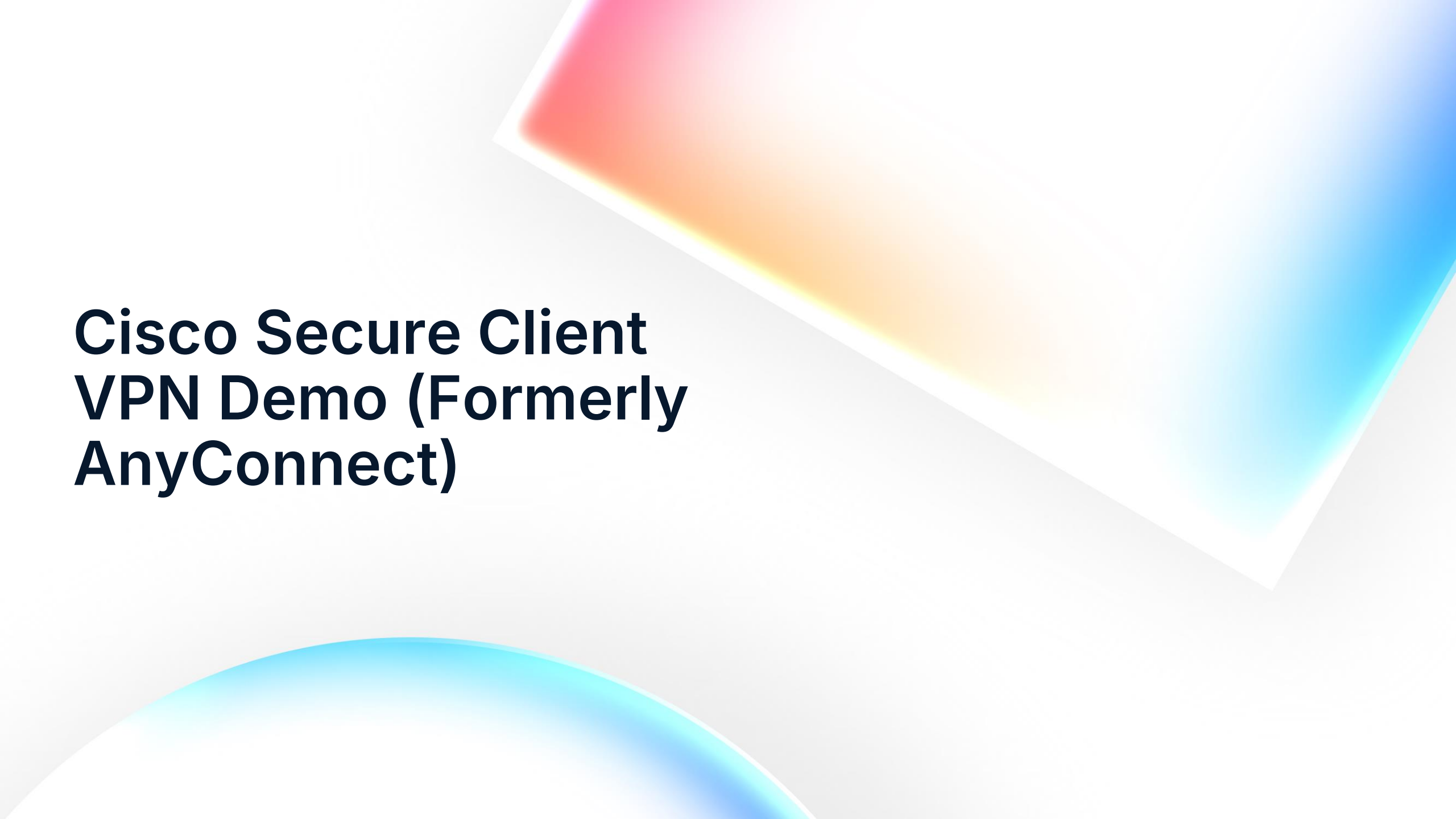
- VPNs provide a bad user experience.
- VPN assumes that anyone or anything passing network perimeter controls can be trusted.
- ZTNA (Zero Trust Network Access) takes the opposite approach by not trusting any user or device until proven otherwise.
- ZTNA extends the zero-trust model beyond the network.
- ZTNA reduces the attack surface by hiding applications from the internet



Although traditional VPNs have been a mainstay for decades, ZTNA is the natural evolution of VPN and offers better security, more granular control, and a better user experience in light of the complexity of today's networks, so it can be a smarter choice for securely connecting a remote workforce.

Zero Trust, ZTA, and ZTNA: What's the difference? - CSO

<https://www.csoonline.com/article/570475/zero-trust-zta-and-ztna-what-s-the-difference.html>



Cisco Secure Client VPN Demo (Formerly AnyConnect)



Macintosh HD



VPN Objections

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VPN Objections

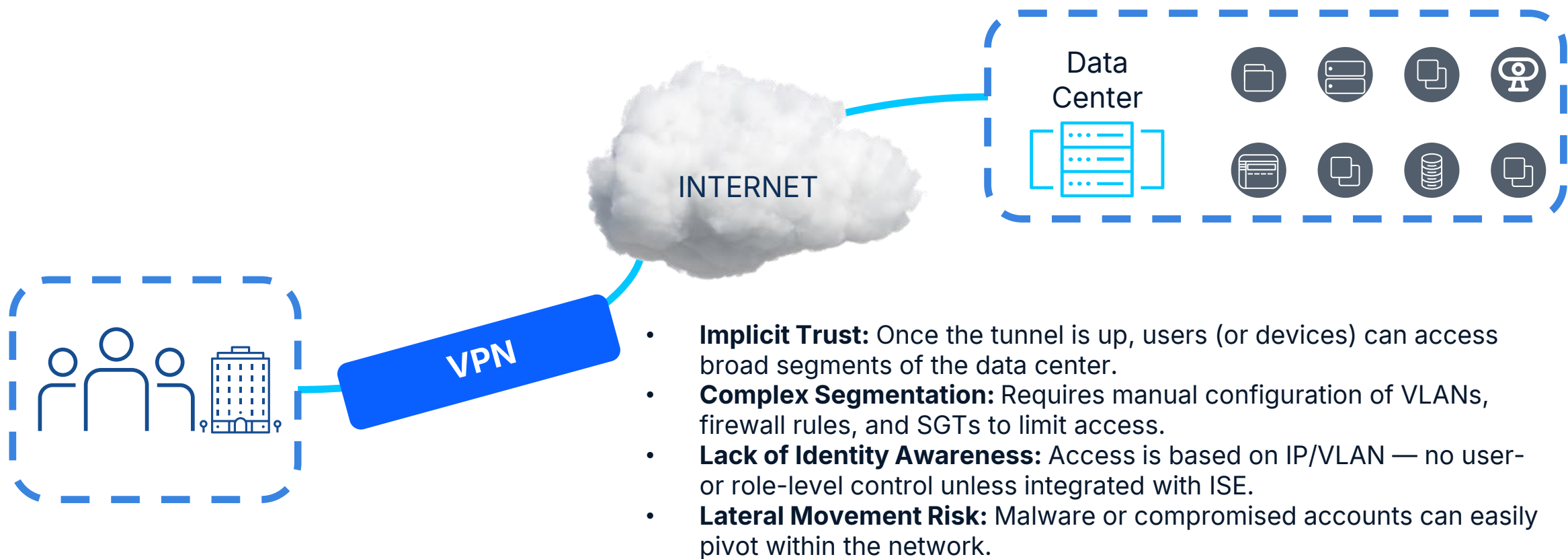
- VPNs provide a bad user experience.
- VPN assumes that anyone or anything passing network perimeter controls can be trusted.
- ZTNA (Zero Trust Network Access) is a more secure approach by not trusting any user or device until proven otherwise.
- ZTNA extends the zero trust model beyond the network.
- ZTNA reduces the attack surface by hiding applications from the internet

BUSTED

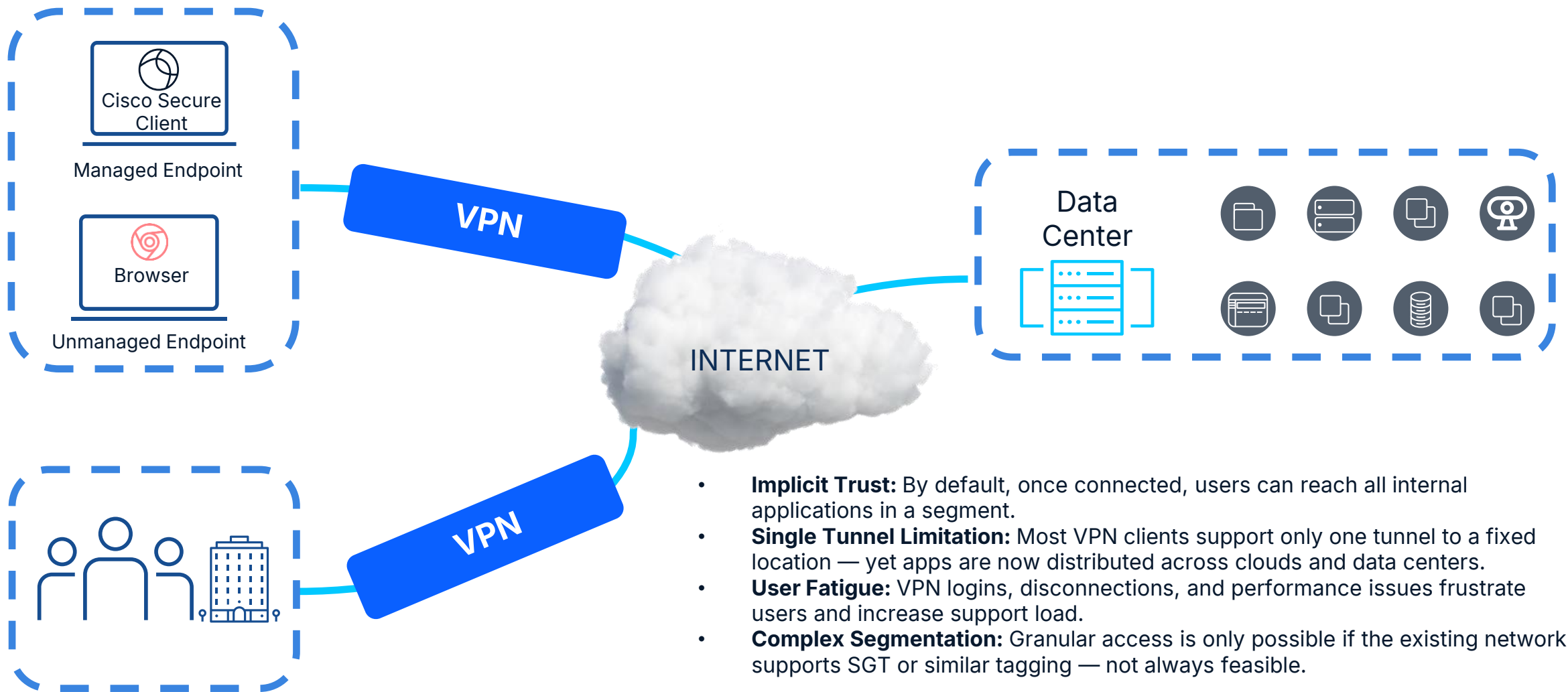
Why Zero Trust Network Access (ZTNA)?

The background features a light gray grid pattern. Overlaid on this are several large, semi-transparent geometric shapes. A prominent shape in the upper right is a parallelogram with a color gradient from red to orange to yellow. Another shape in the lower right is a parallelogram with a blue to cyan gradient. In the bottom left, there is a large, curved shape with a blue to cyan gradient.

Users in Branch accessing Apps in DC

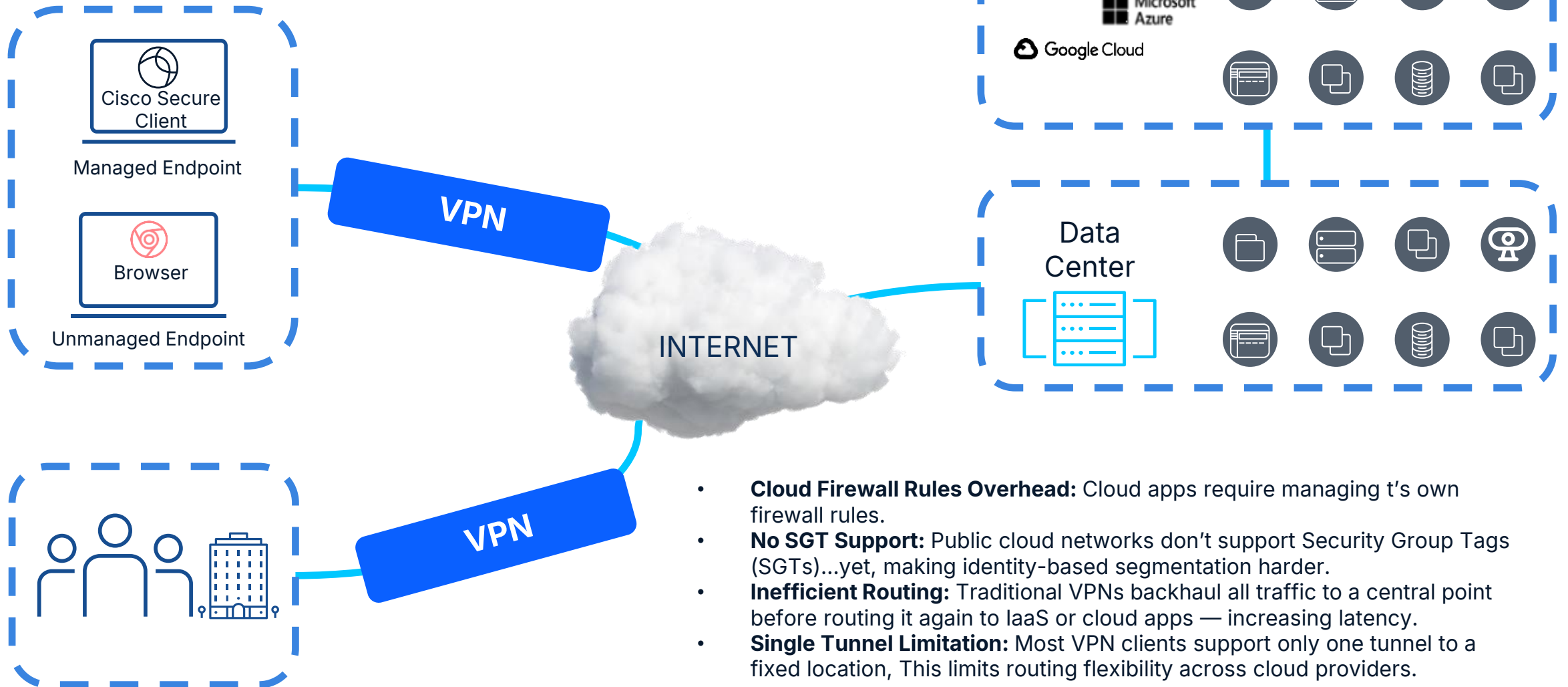


Now Add Remote Users



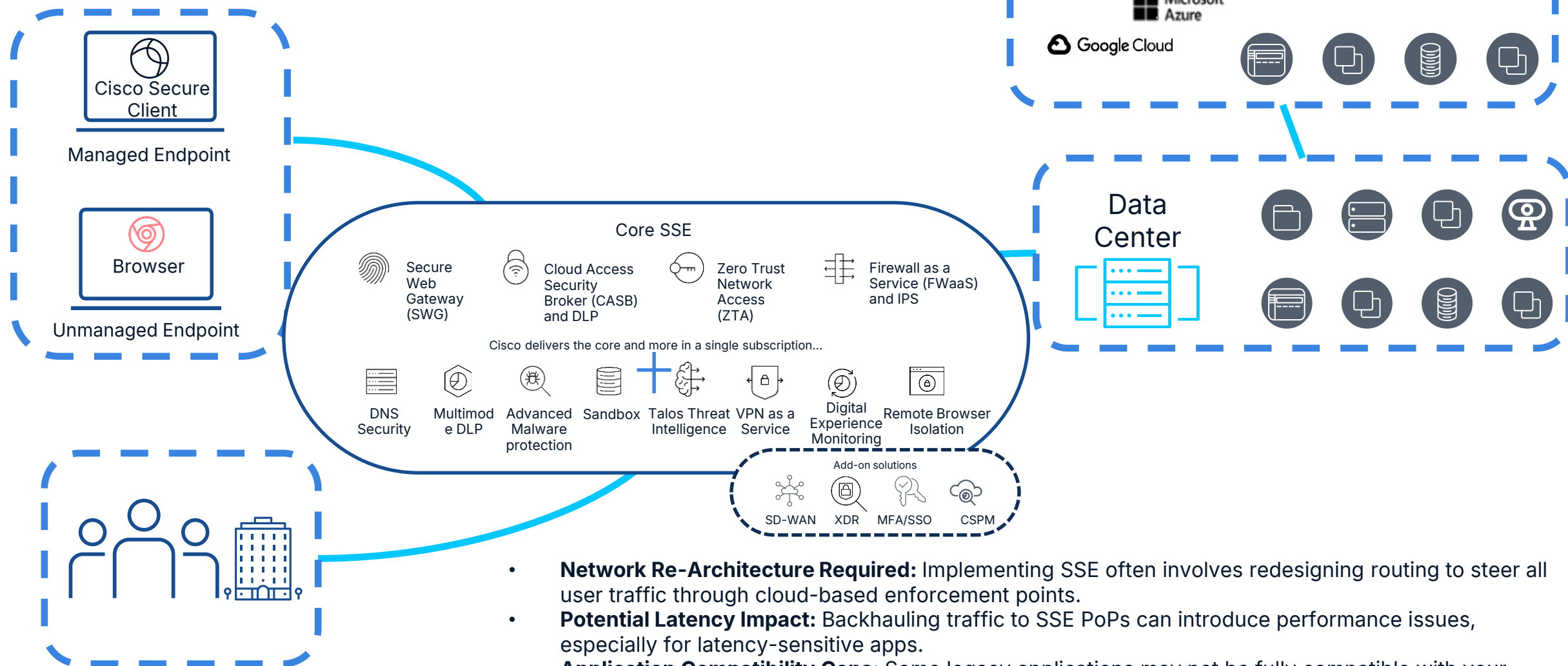
- **Implicit Trust:** By default, once connected, users can reach all internal applications in a segment.
- **Single Tunnel Limitation:** Most VPN clients support only one tunnel to a fixed location — yet apps are now distributed across clouds and data centers.
- **User Fatigue:** VPN logins, disconnections, and performance issues frustrate users and increase support load.
- **Complex Segmentation:** Granular access is only possible if the existing network supports SGT or similar tagging — not always feasible.

Then Add Apps in the Cloud



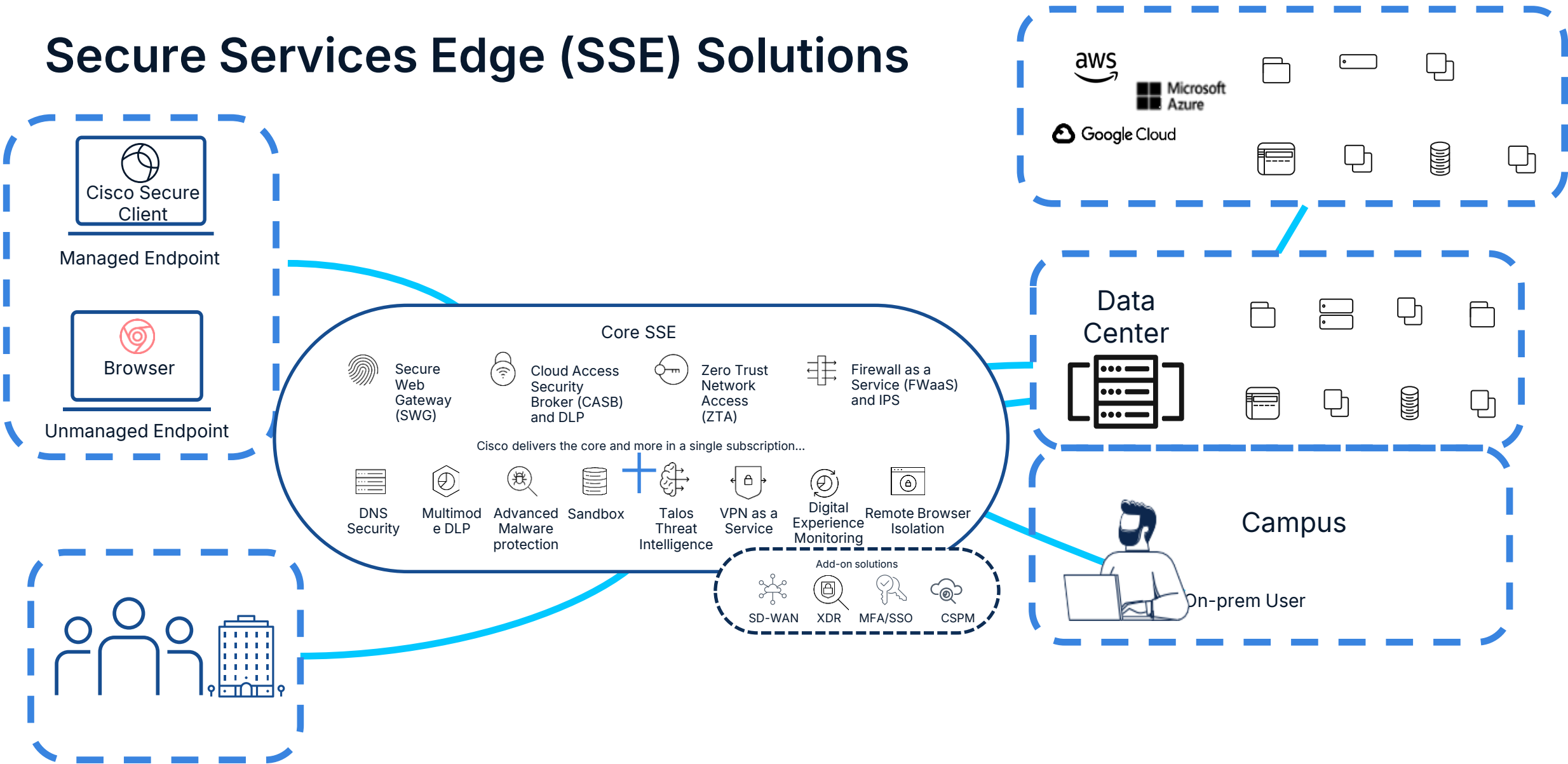
- **Cloud Firewall Rules Overhead:** Cloud apps require managing t's own firewall rules.
- **No SGT Support:** Public cloud networks don't support Security Group Tags (SGTs)...yet, making identity-based segmentation harder.
- **Inefficient Routing:** Traditional VPNs backhaul all traffic to a central point before routing it again to IaaS or cloud apps — increasing latency.
- **Single Tunnel Limitation:** Most VPN clients support only one tunnel to a fixed location, This limits routing flexibility across cloud providers.

Secure Services Edge (SSE) Solutions



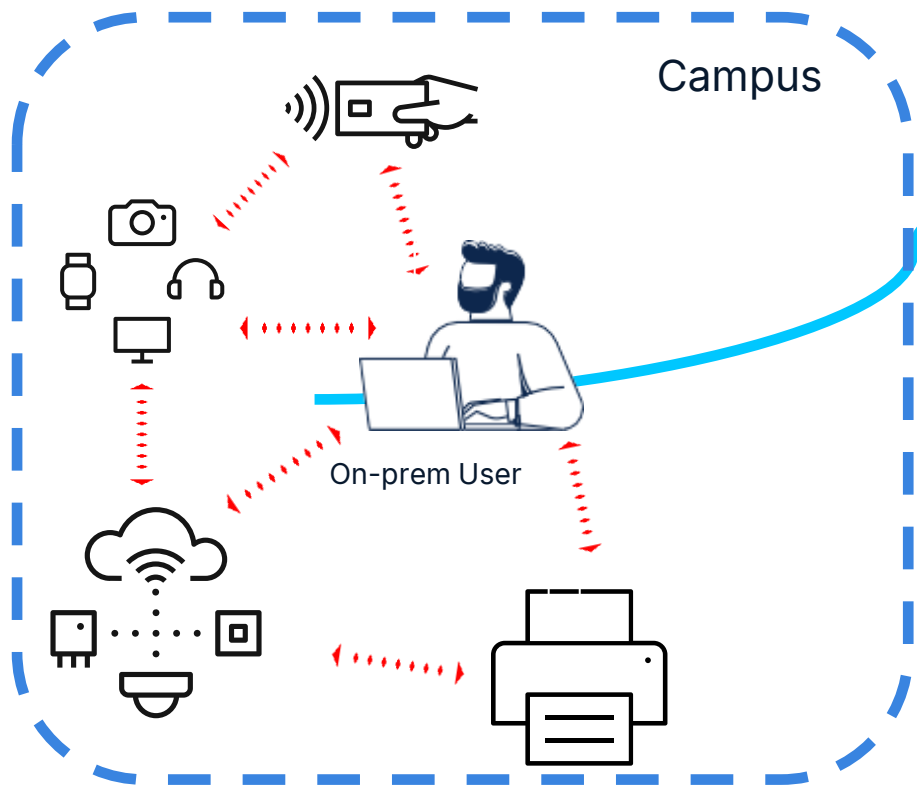
- **Network Re-Architecture Required:** Implementing SSE often involves redesigning routing to steer all user traffic through cloud-based enforcement points.
- **Potential Latency Impact:** Backhauling traffic to SSE PoPs can introduce performance issues, especially for latency-sensitive apps.
- **Application Compatibility Gaps:** Some legacy applications may not be fully compatible with your chosen ZTNA solution.
- **Operational Complexity:** Troubleshooting user issues can be more complex than with legacy VPNs.

Secure Services Edge (SSE) Solutions

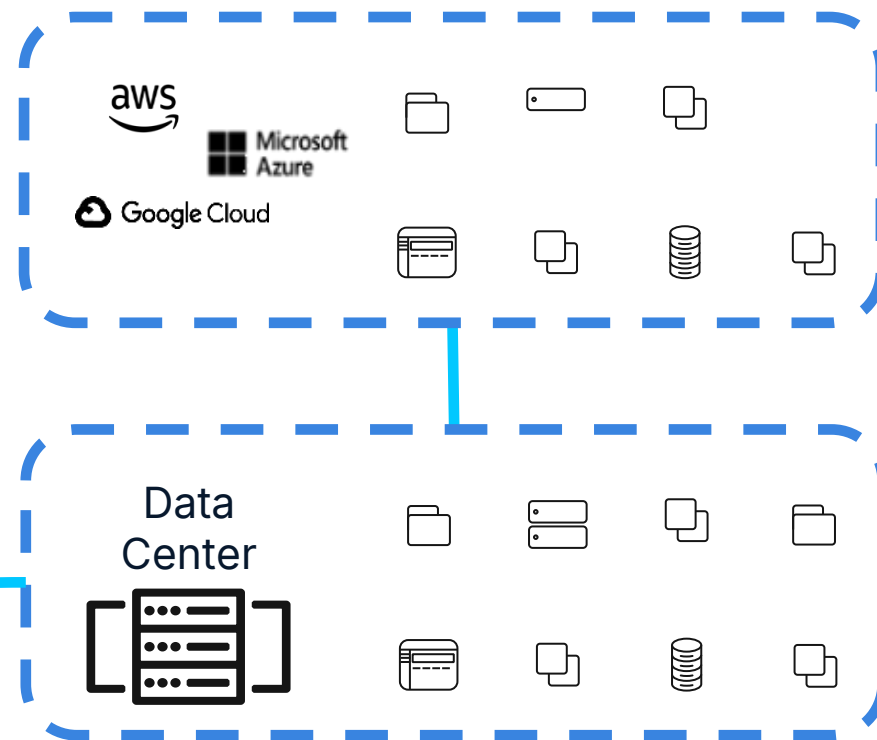


Does ZTNA solve everything?

- What about non-user devices and things?
- How to segment corporate network?
- What About Legacy Apps?

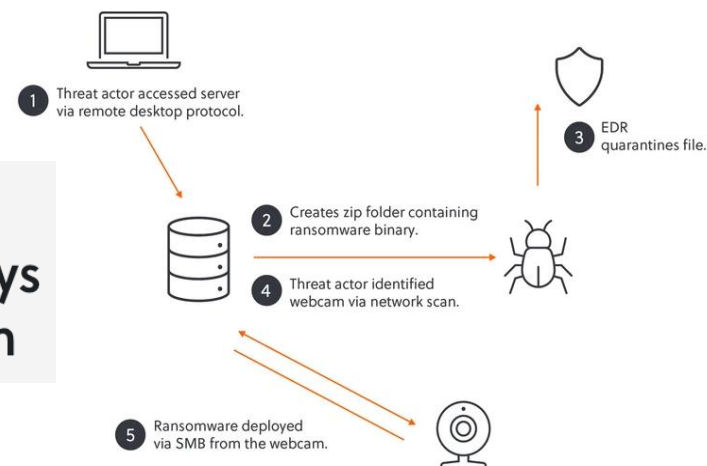


ZTNA



5 March 2025 5 min read

Camera off: Akira deploys ransomware via webcam



<https://www.s-rminform.com/latest-thinking/camera-off-akira-deploys-ransomware-via-webcam>

Universal ZTNA

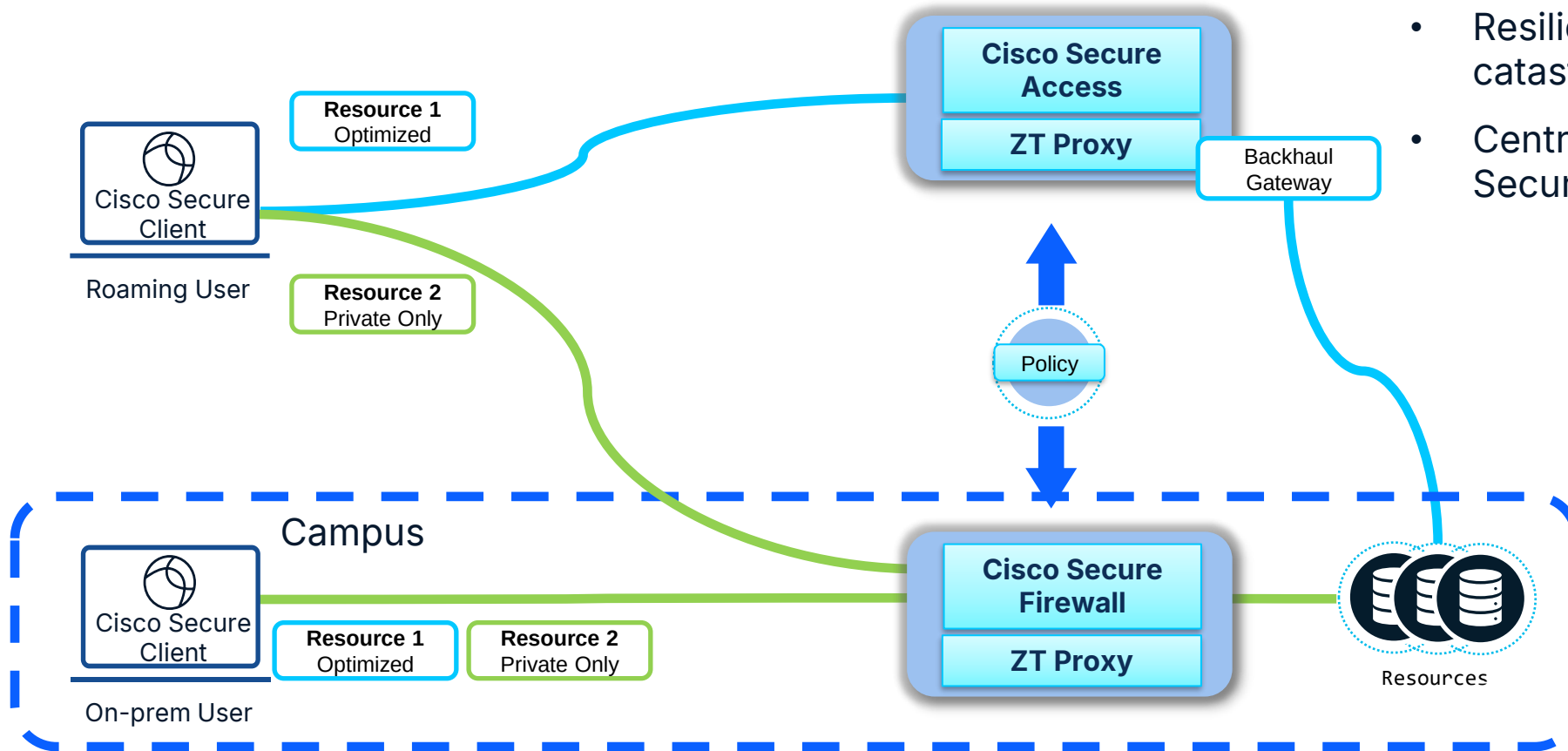


"Having partly started as a "next generation" replacement for VPN connectivity for remote secure access with ZTNA, uZTNA has shifted the value proposition to support local enforcement in on campus and branch "on-premises" locations."

Gartner Emerging Tech Horizon for Communications – April 2025

Cisco Universal ZTNA

- Single Policy, Distributed Enforcement



- Optimal connectivity based on user location.
- Enterprise privacy for sensitive resources.
- Resiliency in case of catastrophic Cloud Outage.
- Centrally managed through Security Cloud Control.

Universal ZTNA Admin Experience



Assign roles

Create organization users and assign roles to control access.

Assign

Other tasks

 CLAIM SUBSCRIPTION

Claim subscriptions to activate instances in your organization.

Claim

SET YOUR DEFAULT LANDING PAGE

Select a product page to view first when you enter the organization.

Select

ONBOARD FIREWALL DEVICES

Manage your firewall in Security Cloud Control.

Onboard

 INTEGRATE IDENTITY PROVIDER (IDP)

Provide single sign-on (SSO) to your organization's users.

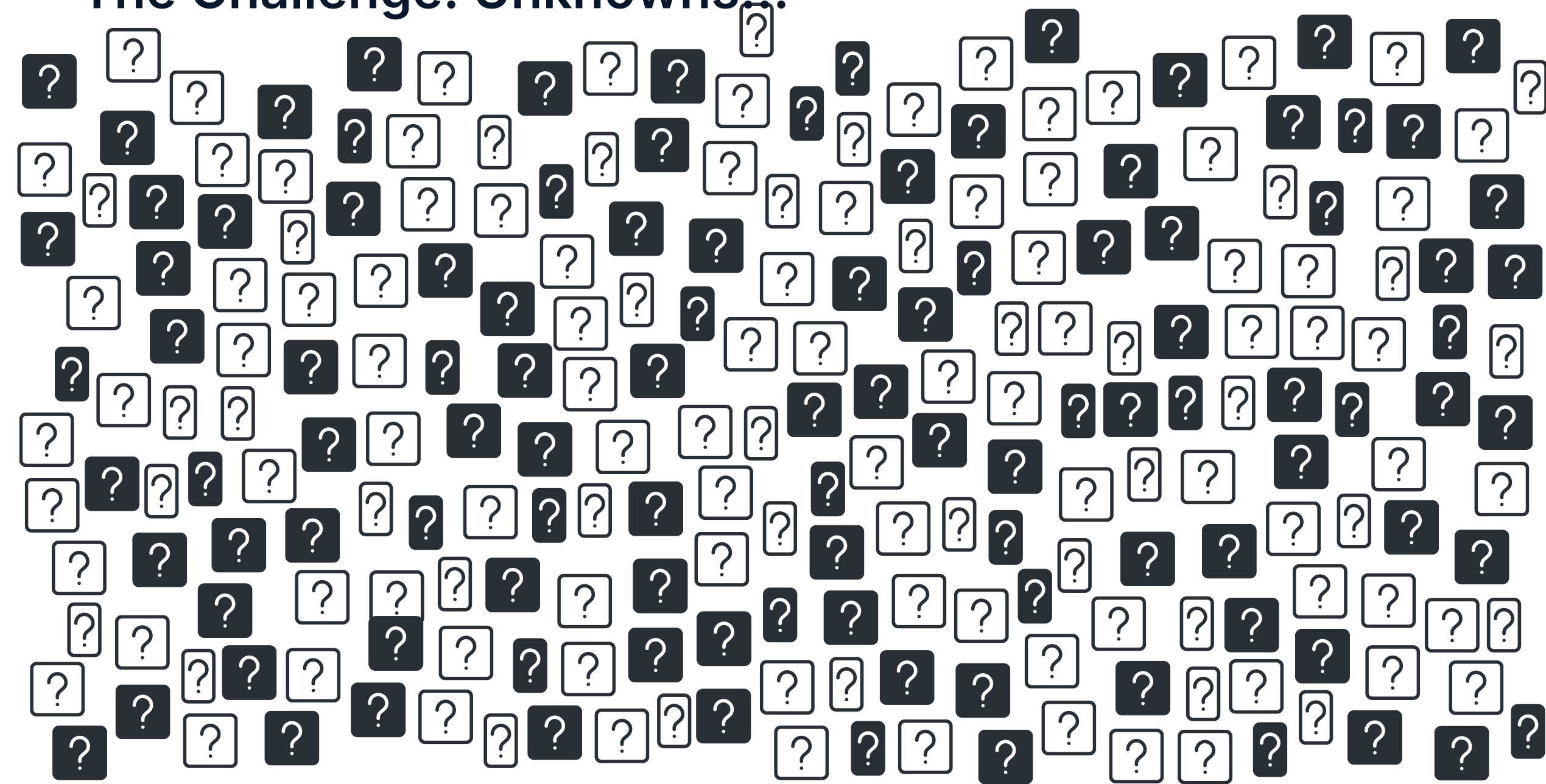
Configure

Universal Identity makes UZTNA Possible



There is no universal zero trust without ubiquitous, shared identity across the enterprise

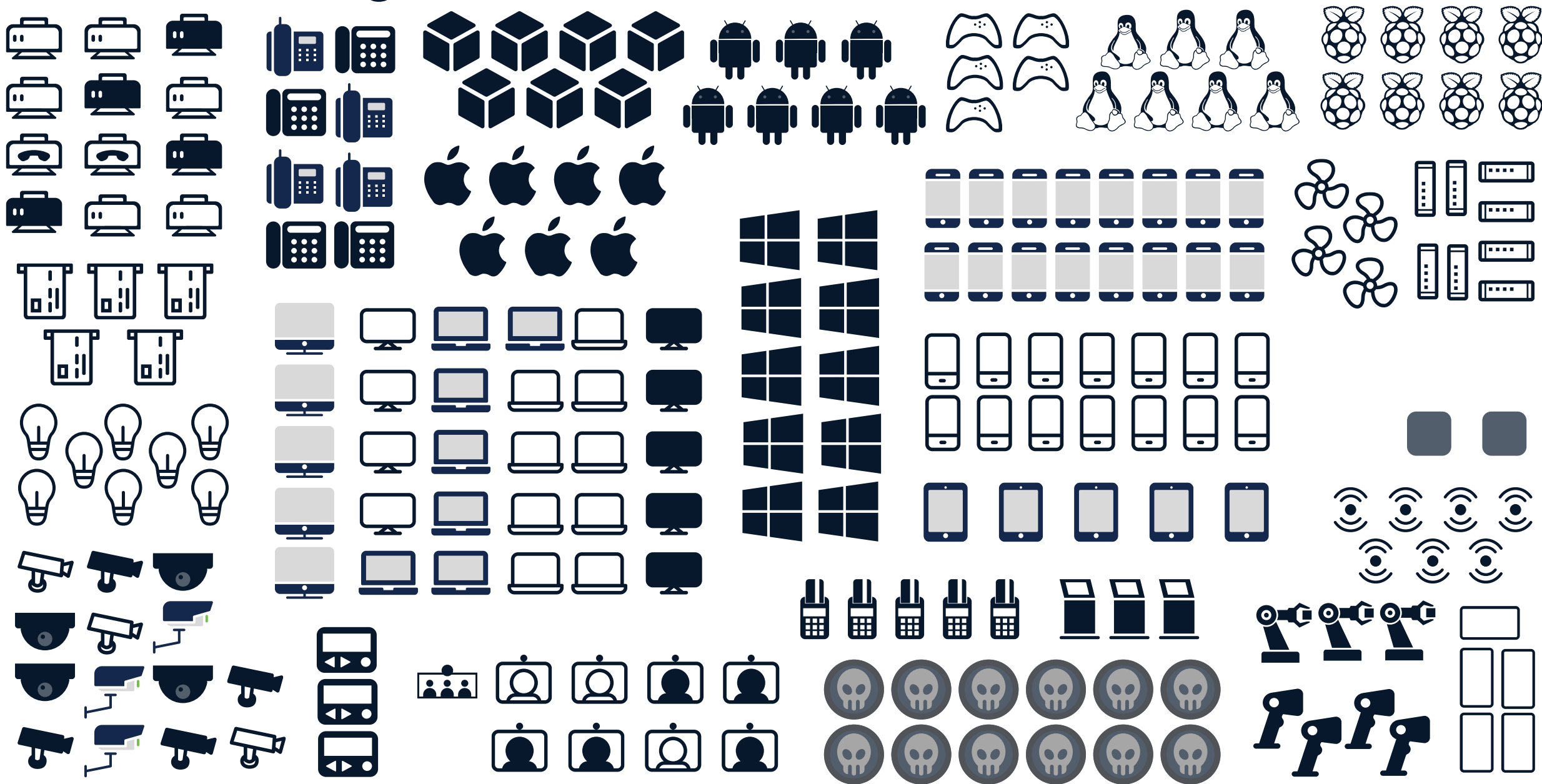
The Challenge: Unknowns.



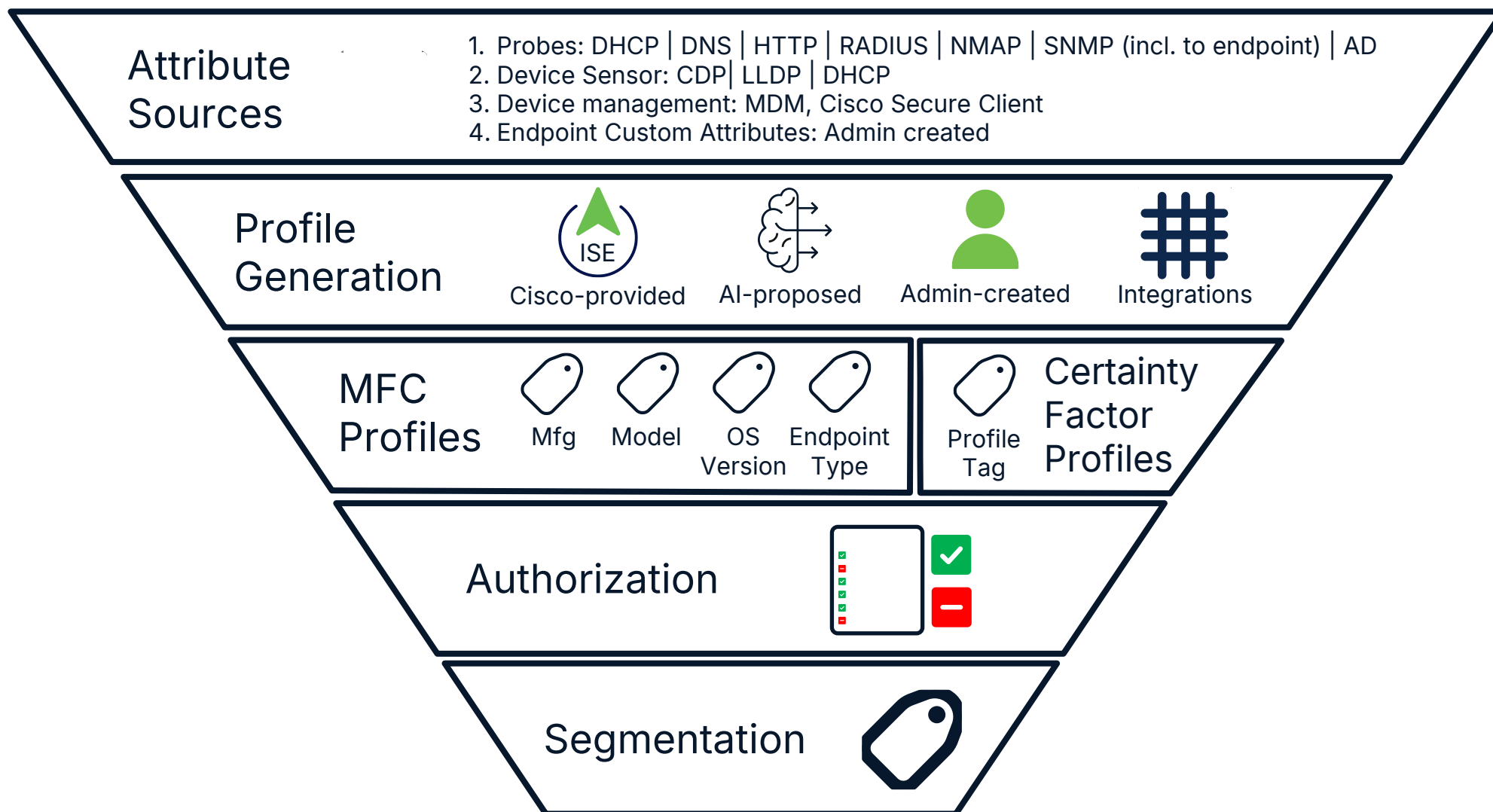
The Challenge: Unknowns to Known...



The Challenge: Unknowns to Known to Classified



Turning Attributes Into Profiles, Profiles Into Protection

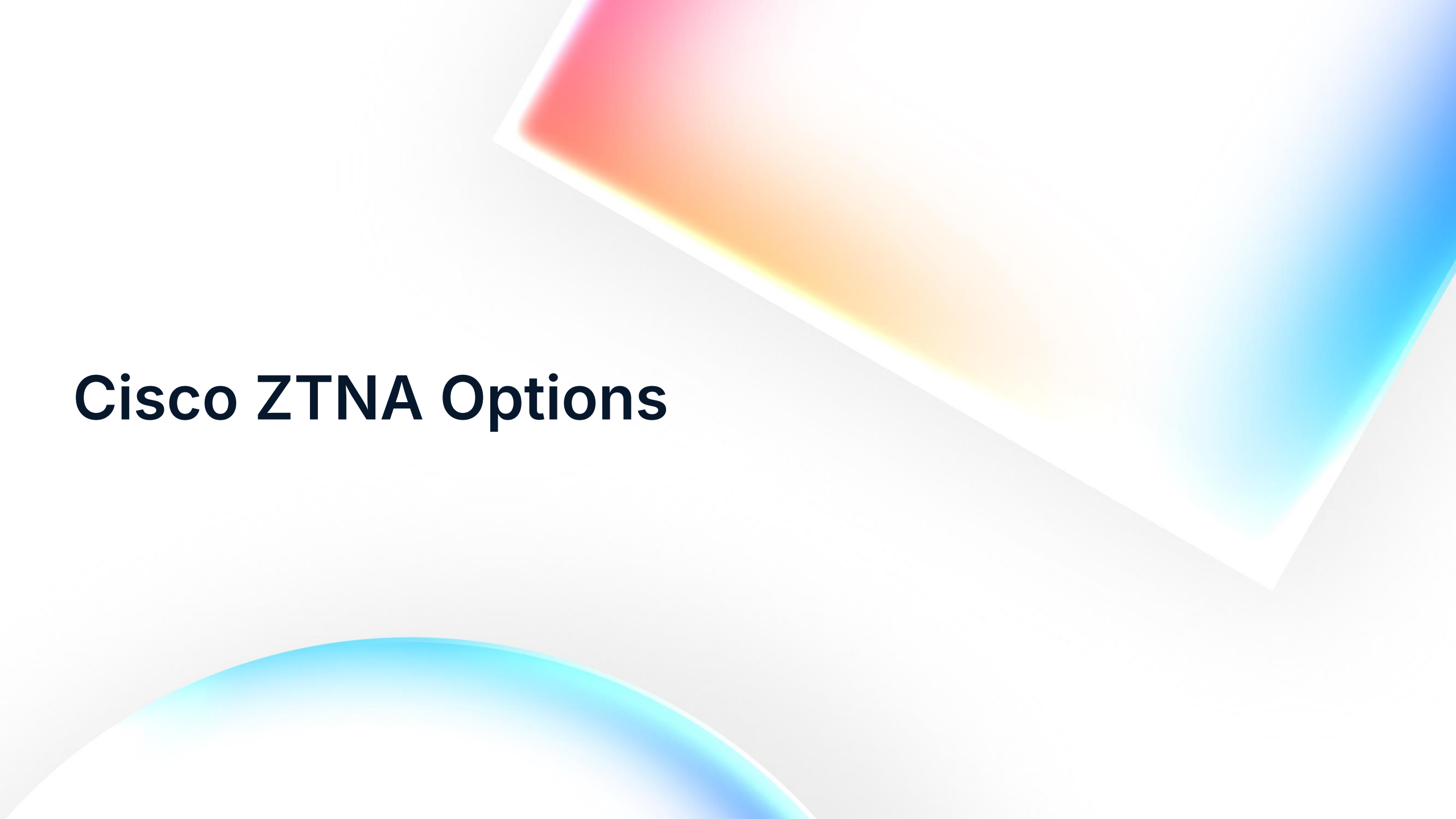


Security Group Tags (SGTs) based policy

Provides SGT attributes from ISE as a source in policy in Secure Access

The image displays three overlapping screenshots of the Cisco Secure Access web interface, illustrating the steps to configure a Security Group Tags (SGT) based policy. The screenshots are numbered 1, 2, and 3.

- Screenshot 1:** Shows the 'Integrations' page. A sidebar on the left contains navigation links: Home, Experience Insights, Connect, Resources, Secure, Monitor, Admin, and Workflows. The main content area is titled 'Cisco Identity Services Engine' and provides instructions on integrating Secure Access with ISE.
- Screenshot 2:** Shows the 'Security Group Tags' page. It features a search bar and a list of SGTs with columns for Name and Count. The list includes: ANY, AP_EMR_EPG, AP_Services_EPG, AP_Test_EPG, Auditors, BYOD, Cameras, Contractors, Developers, and Development_Servers.
- Screenshot 3:** Shows the 'Specify Access' configuration page for a rule named 'VideoEquipmentAccess' with a rule order of 15. The 'Action' section has two options: 'Allow' (selected) and 'Block'. The 'From' section is for specifying sources, showing a list of SGTs: BYOD (15), Cameras (24), Contractors (5), and Developers (8). The 'To' section is for specifying destinations, showing 'Private Resources' (5) and 'Private Resource Groups' (2). A 'Save' button is visible in the bottom right corner.

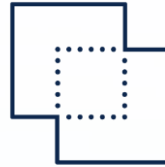


Cisco ZTNA Options

Cisco ZTNA Options



**Duo Network Gateway
(DNG)**



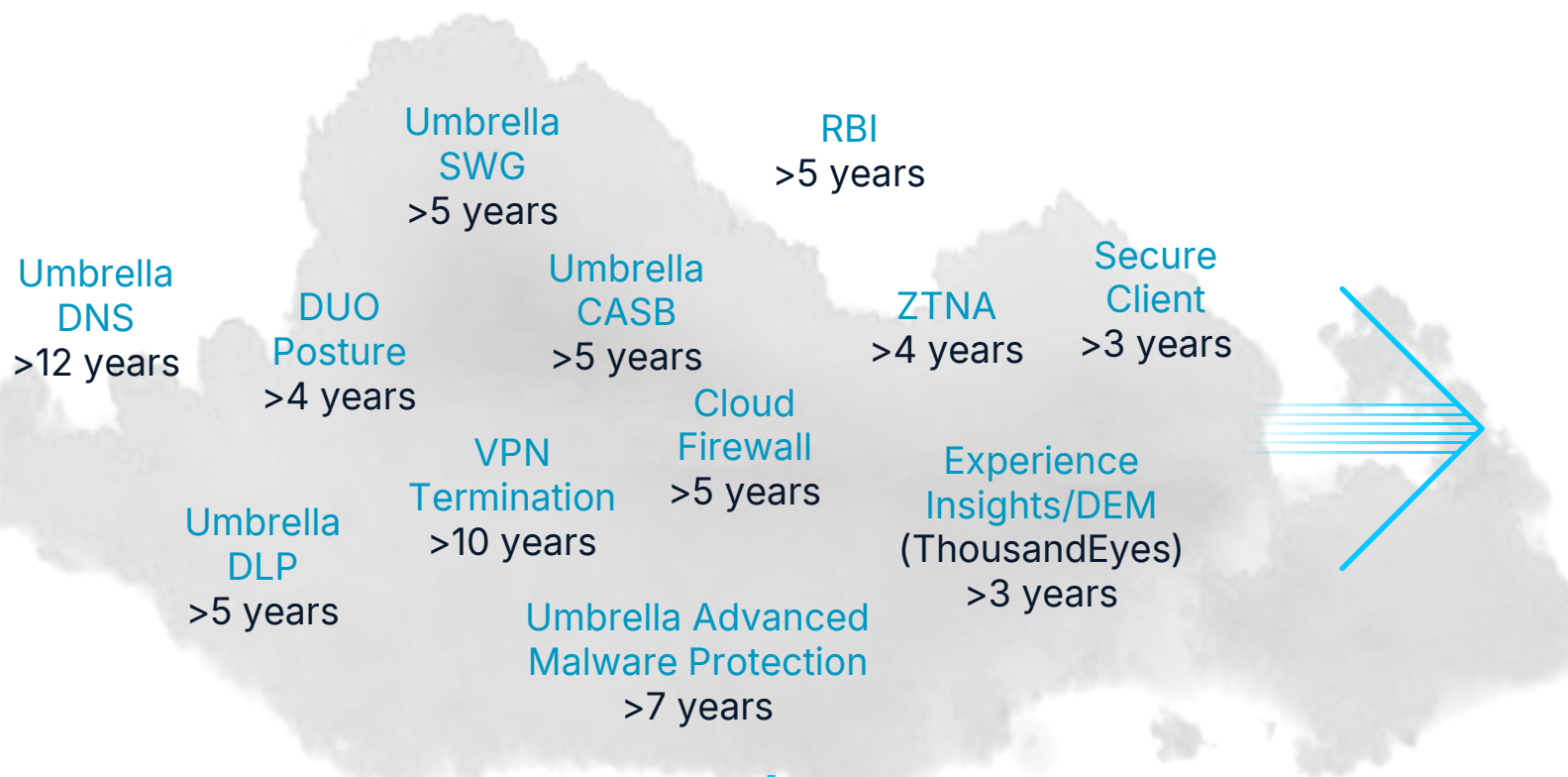
**Secure Firewall
Since 7.4**



Secure Access

Cisco Secure Access

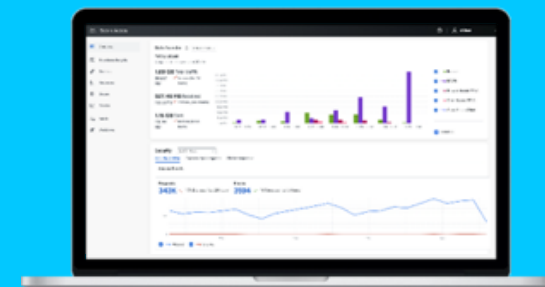
Proven cloud-native security converged into one service



Protecting 70,000+ customers | More than 220M endpoints

Validated by Miercom in their 2024 SSE Benchmark Report

Cisco Secure Access Single EXPERIENCE



- Single Console
- Single Policy
- Single Client
- Single License



Single Client: All-inclusive modules and licensing

AnyConnect VPN (Core)

ZTA Module (new!)

ISE Posture

Secure Endpoint (AMP)

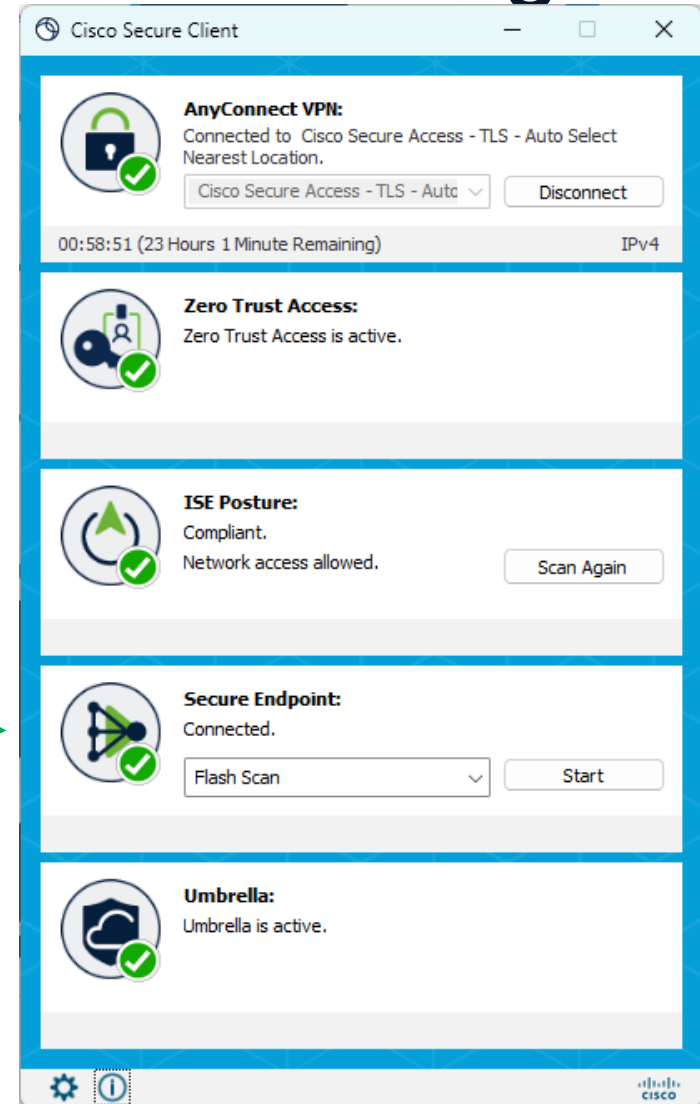
Roaming Module

Thousand Eyes (No UI)

Duo Health (Posture)

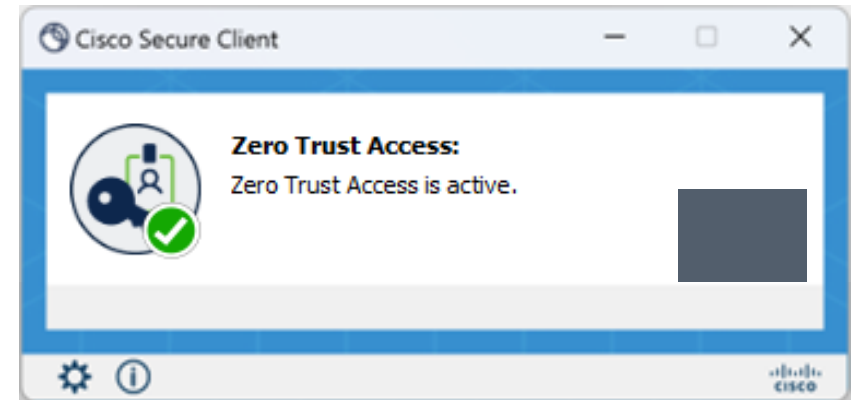
Diagnostic and Reporting (DART)

Cloud Management Module (No UI)



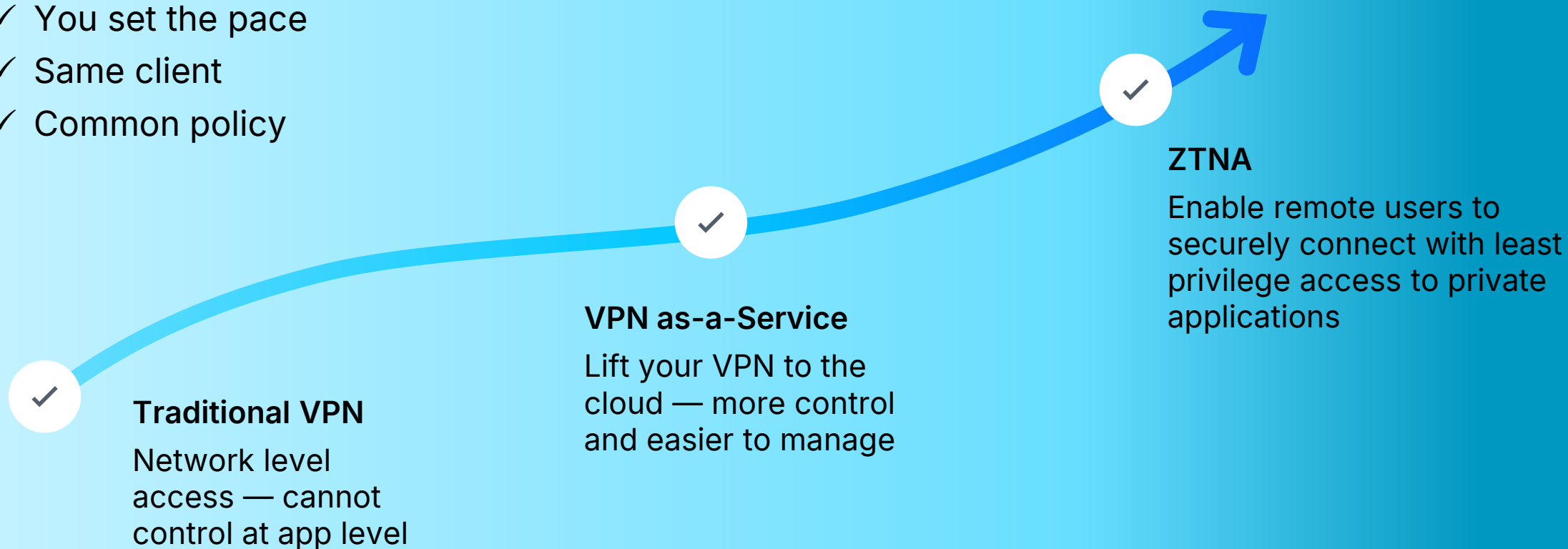
Zero Trust Access Module

- Transparent user experience.
- Proxied resource access with coarse-grained or fine-grained access control.
- Service managed client certificates with TPM/hardware enclave key storage.
- Support for both TCP and UDP applications.
- Cisco and third-party VPN client interop.
- Next-generation protocol (QUIC & MASQUE).



Flexible Journey to ZTNA

- ✓ You set the pace
- ✓ Same client
- ✓ Common policy





Cisco Secure Client VPNaaS and Zero Trust Access User Experience Demo

Cisco IT's Journey to SSE

FULL CONFERENCE

IT LEADERSHIP

Cisco IT's Journey to SSE - Enabling Hybrid Work with Cisco Secure Access - BRKCOC-2040



Mike Cifelli, Network Systems Engineer, Cisco

Joe DeSanto, IT Technical Leader, Cisco - **Distinguished Speaker**

Schedule

Tuesday, Jun 10 | 4:30 PM - 5:30 PM PDT

Cisco IT's journey to Security Service Edge (SSE) highlights prevailing success in our strategic expansion ensuring consistent, reliable, and secure hybrid worker connectivity worldwide.

Through the integration of VPNaaS, Zero Trust Network Access (ZTNA), and Generative AI protection with Secure Web Gateway (SWG), all managed by the Cisco Secure Access cloud service, customers will learn how our adoption is achieving robust and secure access without the need for hardware.

You will also gain insights into our journey on how we are transforming user experience and security while overcoming IT business challenges and ultimately paving the way for enterprise scale and stability.

Session Type: Breakout

Technical Level: Intermediate

Technology: SASE, Hybrid Work, Security

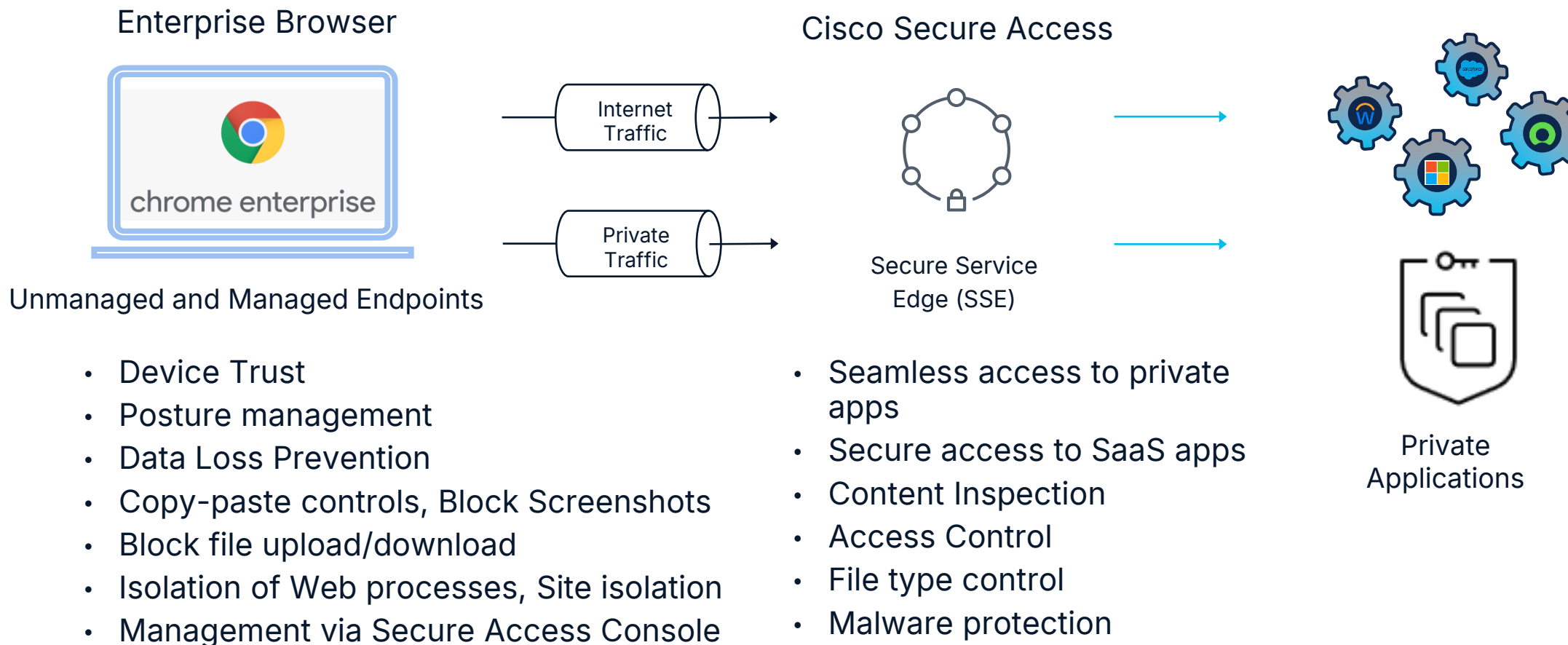
Track: Security



Wipro's Secure Access and Chrome Enterprise Browser Integration

Secure Access with Enterprise Browser

Zero Trust Access to Private Apps and Internet Apps





Key takeaways

Key Takeaways

- **VPNs aren't dead** — but they're no longer enough. While VPNs still serve remote access use cases, they fall short in today's hybrid, cloud-first environments.
- **ZTNA advances the model** — but doesn't solve everything. It brings identity-based access and reduced attack surface, yet struggles with on-prem segmentation, non-user-based devices and legacy applications.
- **UZTNA takes it further** — but isn't a silver bullet. It extends ZTNA to support broader use cases, including on-prem users, but it doesn't replace the advanced visibility and containment features that Cisco ISE provides — the two work best together.
- **Zero Trust isn't a product** — it's a strategy. Choose and combine technologies based on your specific environment, risk tolerance, and operational goals.

Slido



•How many Cisco Lives have you attended?



- The Slido app must be installed on every computer you're presenting from



- Which protocol is used by Cisco Secure Access for Client Based Zero Trust Network Access (ZTNA)?



- Which of the following is a core advantage of Universal ZTNA (UZTNA) over traditional ZTNA?



- The Slido app must be installed on every computer you're presenting from



- **Which feature is a key characteristic of ZTNA but not typically found in VPNs?**



- The Slido app must be installed on every computer you're presenting from



- What is a significant challenge faced in the use case of a local user accessing an application within the same location in a Zero Trust Network Access (ZTNA) environment?



- **Which connectivity options are available in Cisco Secure Access for accessing private applications?**



- Who is the legendary Costa Rican goalkeeper known for starring in multiple World Cups and playing for clubs like Real Madrid and PSG?

Complete your session evaluations



Complete a minimum of 4 session surveys and the Overall Event Survey to be entered in a drawing to win 1 of 5 full conference passes to Cisco Live 2027.



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Attend the interactive education with DevNet, Capture the Flag, and Walk-in Labs



Visit the On-Demand Library for more sessions at www.CiscoLive.com/on-demand

CISCO Live !

Thank you

